Introduction to Causal Inference and Causal Data Science

Day 5: Mediation and Complex Longitudinal Settings

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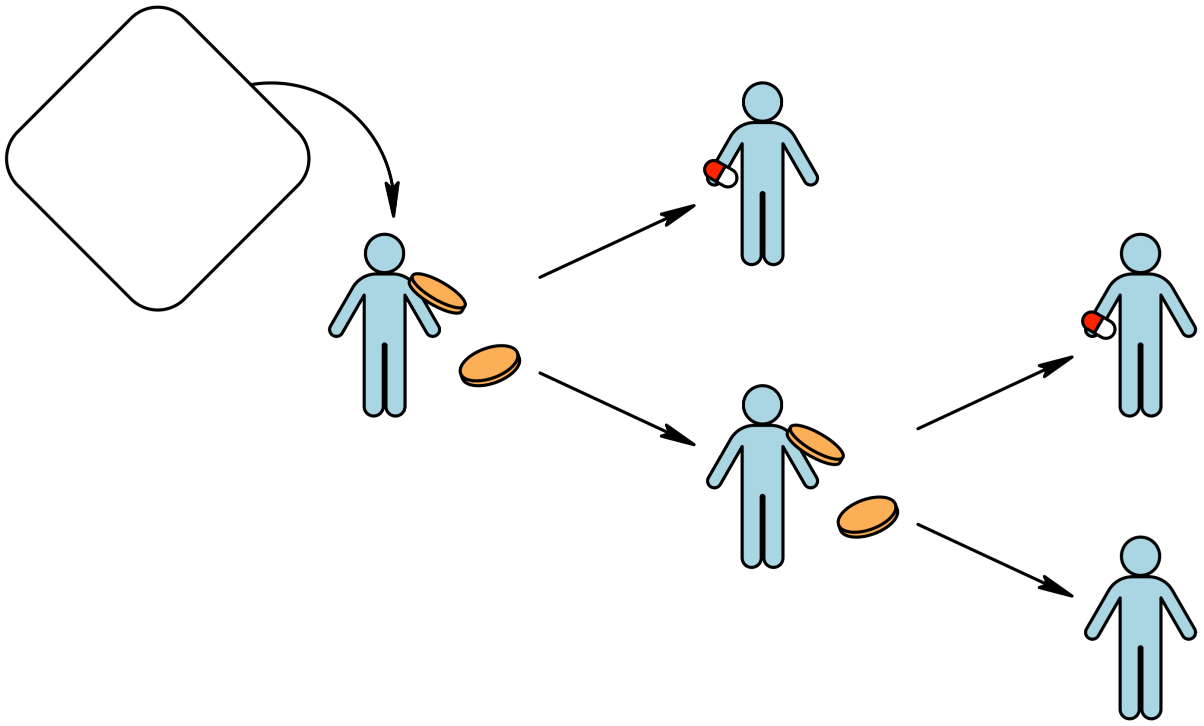
Topics

- A **taxonomy of estimands** on time-varying treatments (incl. intention-to-treat and per-protocol effects) and mediation
- Why/when **traditional methods fail** for inference about time-varying treatments or mediation
- **G-methods** for time-varying treatments and mediation

Inference about time–varying treatments

If treatment/exposure is time-varying, there are **many possible causal contrasts** ...

- single- versus multiple-point interventions
- $2^{\#\text{ time points}}$ static rules
- Static versus dynamic



Single-point (baseline) intervention

Decision at baseline only — individuals are allowed to deviate

- *Assign/initiate* versus withhold drug treatment at baseline (*intention-to-treat*)

Multiple-point (joint) intervention

Decisions/randomisation at multiple points

- Sustained/daily/weekly/monthly drug use versus continuous non-use (*per-protocol*)

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Static treatment rule/regime/protocol/...

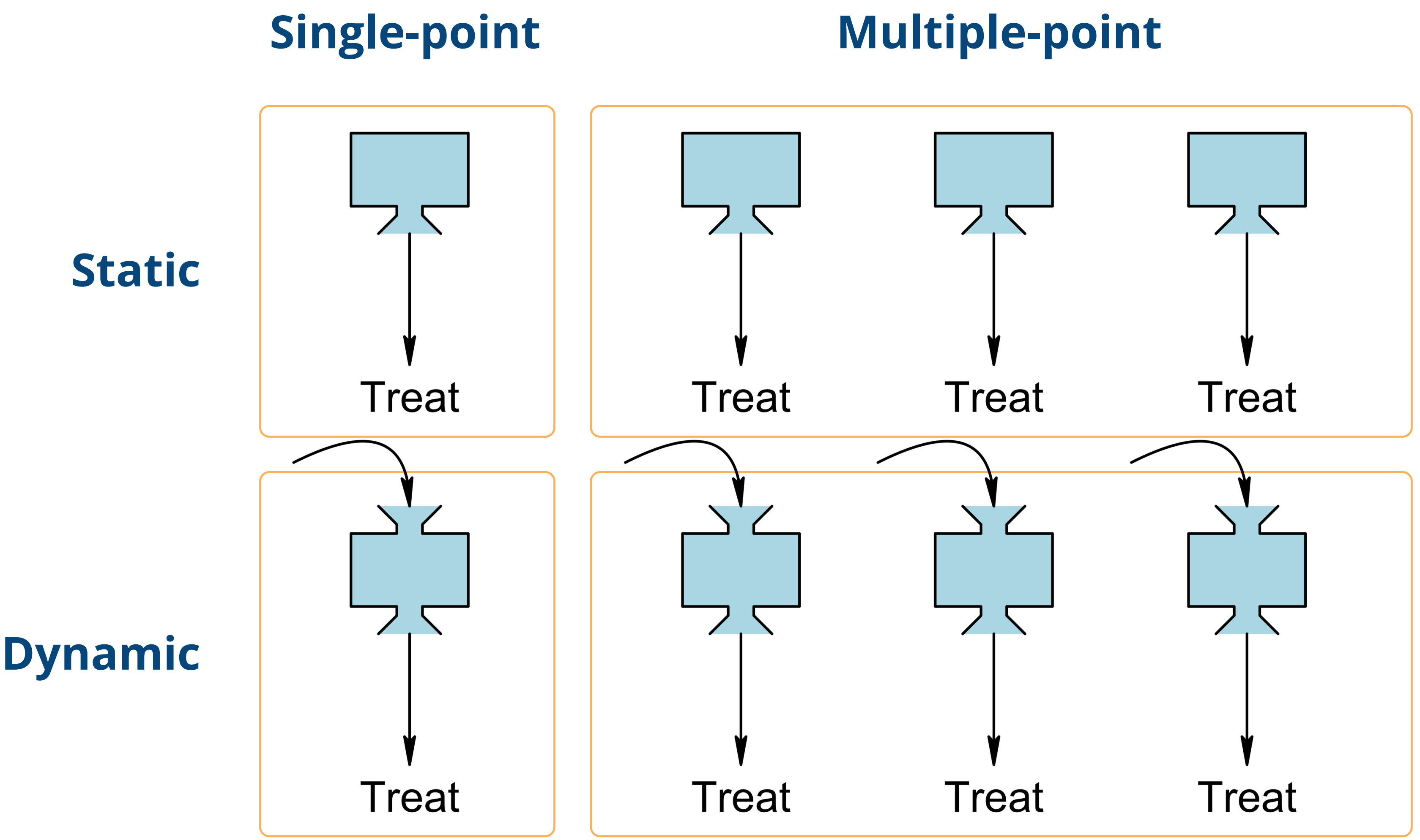
... assigns the same treatment option to everyone

- Assign versus withhold treatment programme at baseline (*intention-to-treat*)
- Always treat versus never treat (*per-protocol*)

Dynamic (individualised) treatment rule

... assigns treatment based on the then-available information

- Choose dose depending on baseline covariates
- Start when blood marker first drops below threshold
- Stop when toxicity occurs



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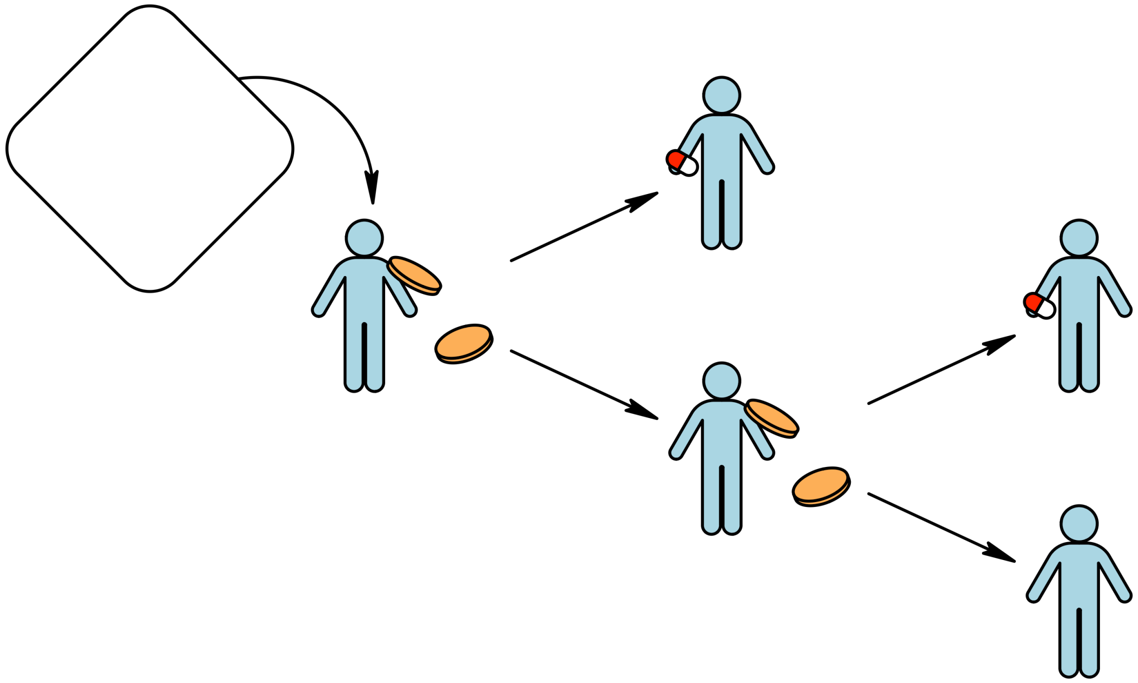
Inference about always-/never-treat strategies

- Trials with *baseline randomisation* naturally answer questions on *point interventions*.
- Questions about *always-/never-treat strategies* are most naturally addressed by trials with (full adherence or) *sequential randomisation*.



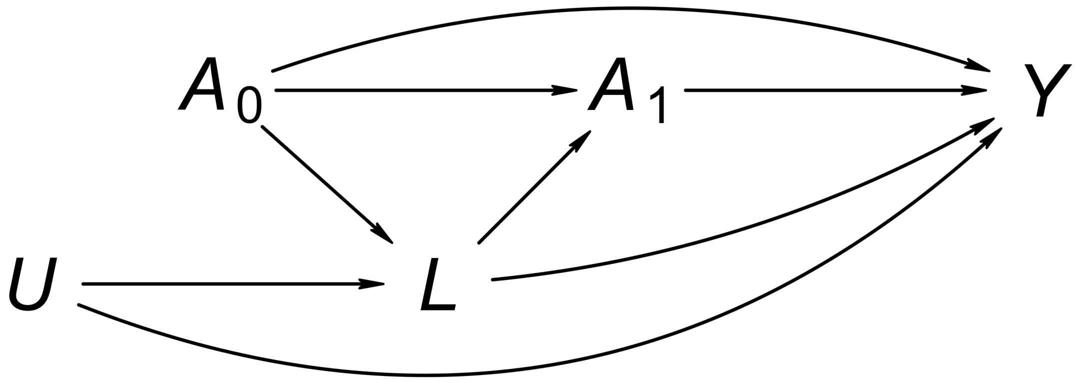
For simplicity, let's assume there are only two intervention times, t_0 and t_1:

- A_0 and A_1 are indiciator variales for treatment at t_0 and t_1, respectively
- Y^{a_0,a_1} is the outcome if, possibly contrary to fact, $A_0 = a_0$ and $A_1 = a_1$

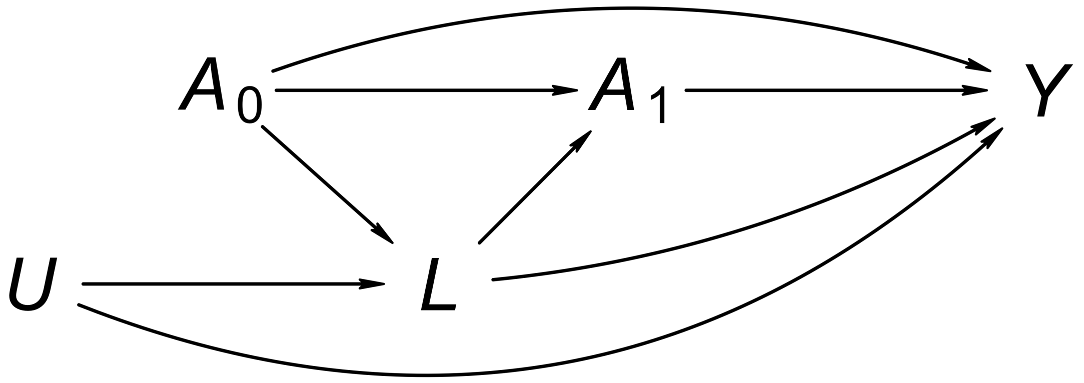
Identification of always- versus never-treat effect (ATE) under sequential randomisation

$$
\begin{aligned}
\text{ATE} &= \mathrm{E}[Y^{1,1}] - \mathrm{E}[Y^{0,0}] && \text{(causal estimand, assuming two intervention times)}\\
&= \mathrm{E}[Y^{1,1} \mid A_0 = 1] - \mathrm{E}[Y^{0,0} \mid A_0 = 0] && \text{(\textit{randomisation} at } t_0\text{)}\\
&= \mathrm{E}[Y^{1,1} \mid A_0 = 1, A_1 = 1] - \mathrm{E}[Y^{0,0} \mid A_0 = 0, A_1 = 0] && \text{(\textit{randomisation} at } t_1\text{)}\\
&= \mathrm{E}[Y \mid A_0 = 1, A_1 = 1] - \mathrm{E}[Y \mid A_0 = 0, A_1 = 0] && \text{(\textit{consistency})}
\end{aligned}
$$

But what if treatment is not assigned by (sequential) randomisation? How to address such departures from target trial?



DAG after Naimi et al. (2017). Here, A_0 may represent anti-retroviral treatment at time 0, L HIV viral load just prior to the second round of treatment, A_1 anti-retroviral treatment status at time 1, Y the CD4 count measured at the end of follow-up, U and an unmeasured common cause of HIV viral load and CD4.



🧩 Condition on ("adjust for") L? Hint: we want to block non-causal paths, while leaving all direct (causal) paths from A_0 or A_1 to Y open!

⚠️ Traditional methods for confounding adjustment

Traditional methods (multivariable regression modelling) are *not suited* for time-varying confounding adjustment when there is *covariate-treatment feedback*!

Methods that can handle treatment-covariate feedback and adjust for time-varying confounding:

- *IPW*
- G-computation
- G-estimation

IPW for inference about sustained treatment strategies

If, in addition to consistency and positivity, there is *sequential conditional exchangeability* of the form

$$Y^{a_0,a_1} \perp\!\!\!\perp A_0 \mid L_0 \text{ and } Y^{a_0,a_1} \perp\!\!\!\perp A_1 \mid L_0, L_1, A_0{=}a_0$$

(i.e., L_0 blocks all backdoor paths from A_0 to Y, and L_0, L_1, A_0 together block all backdoor paths from from A_1 to Y), then

$$\mathrm{E}[Y^{a_0,a_1}] = \mathrm{E}_{\text{pseudopopulation}}[Y \mid A_0{=}a_0, A_1{=}a_1],$$

where the pseudopopulation is obtained by weighting everyone by $W_0 W_1$ with

- $W_0 = A_0 \,/\, \Pr(A_0 = 1 \mid L_0) + (1 - A_0) \,/\, \Pr(A_0 = 0 \mid L_0)$
- and $W_1 = A_1 \,/\, \Pr(A_1 = 1 \mid L_0, L_1, A_0) + (1 - A_1) \,/\, \Pr(A_1 = 0 \mid L_0, L_1, A_0)$

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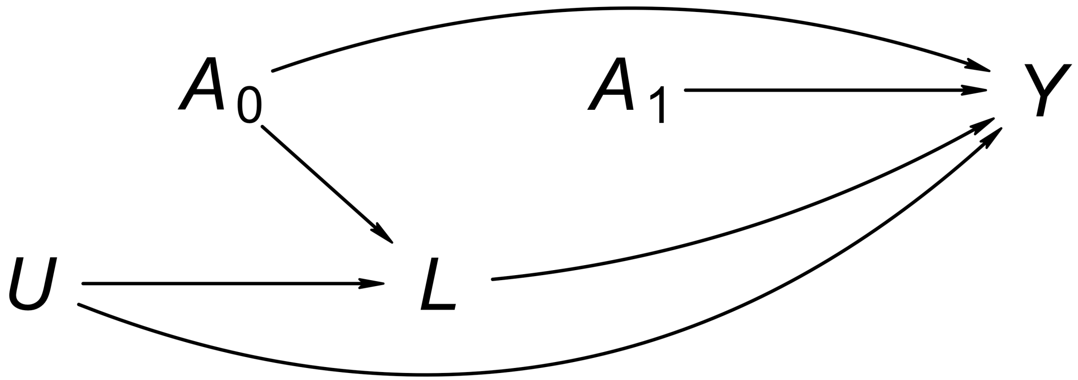
💡 IPW doesn't adjust by conditioning — think of it as surgically removing arrows!



Summary

- IPW and other *g-methods*, but not traditional methods, are suited to handle time-varying confounding affected by past treatment (*feedback*)
- As with IPW for time-fixed confounding, *default standard error estimators* of many software packages are *inappropriate* for IPW regressions, because the weights are falsely assumed to reflect actual (indpendent!) observation frequenties!
- IPW can (and sometimes needs) to be combined with *marginal structural modelling*.

Mediation

What part of the effect “**goes via**” or is “**mediated** by” a third variable?

$$\begin{tikzcd} A \arrow[rr] \arrow[dr] & & Y \\ & M \arrow[ur] & \end{tikzcd}$$

- What part of the effect of *cognitive behavioural therapy (CBT)* on *depression symptoms* is **mediated** by *antidepressant use*?
- To what extent is the effect of *genetic variants* on incident *lung cancer* **mediated** by *smoking behaviour*?
- To what extent is the effect of *maternal smoking* on *infant mortality* **mediated** by *birth weight*?

Notation

- Treatment (e.g., CBT): A
- Mediator (e.g., antidepressant use): M
- Mediator had A been a: M^a
- Outcome (e.g., depression symptoms): Y
- Outcome had $A = a$ and $M = m$: $Y^{a,m}$

$$
\begin{tikzcd}
A \arrow[rr] \arrow[dr] & & Y \\
& M \arrow[ur] &
\end{tikzcd}
$$

In this notation, the counterfactual outcome under a point intervention on A is written Y^{a,M^a}

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Mediation analyis presumes some sort of **decomposition** of the **total**effect into **direct** effects and **indirect** effect:

$$\text{Total effect} = E[Y^{1,M^1} - Y^{0,M^0}]$$
$$= E[Y^{1,M^1} - Y^{1,M^0} + Y^{1,M^0} - Y^{0,M^0}]$$

Natural direct and indirect effects

- The **natural indirect effect**: $E[Y^{1,M^1} - Y^{1,M^0}]$
- The **natural direct effect**: $E[Y^{1,M^0} - Y^{0,M^0}]$

Difficult interpretation! Y^{1,M^0} is a **cross-world counterfactual**!

Instead of *natural* direct effects, we can also look at *controlled* direct effects:

Controlled direct effect

Controlled direct effect (m): $E[Y^{1,m} - Y^{1,m}]$

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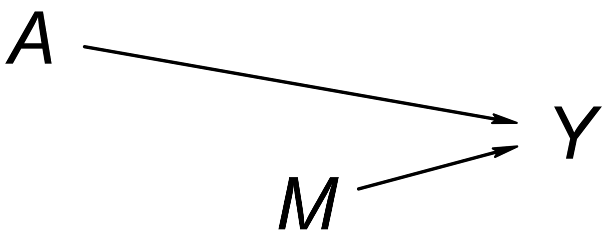
Natural versus controlled direct effects

- Conceptually different
  - NDE: “How much of the effect *flows through* the mediator path?”
  - CDE: “How much of the effect would be removed/preserved by *fixing* the mediator to a constant?”

  CDE more interesting from policy-making perspective!
- Possibly numerically different

We’ve discussed diferent estimands, but
we’ve yet to look into identification/estimation ...



$$
\begin{tikzcd}
A \arrow[rr] \arrow[dr] & & Y \\
& M \arrow[ur] &
\end{tikzcd}
$$

Traditional approach to mediation analysis

- Consists of fitting two linear regression models:
  - First, linearly regress Y on A only — under trial-like conditions (DAG), the *coefficient for A* can be interpreted as *total effect*
  - Second, linearly regress Y on A and M — idea: block path through M; the *coefficient for A* is interpreted as *direct effect*, which is *valid under following model …*

Model underlying traditional approach

$Y^{a,m} = \alpha_0 + \alpha_1 a + \alpha_2 m + U$, for all a and m, with mean-zero $U \perp\!\!\!\perp (A, M)$

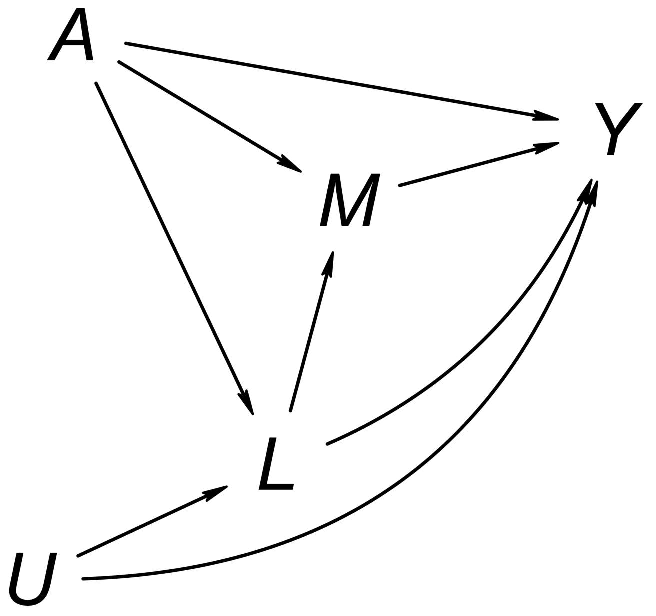
(N.B.: since U is used for all a,m, the individual effects $Y^{1,m} - Y^{0,m}$ are constant — **strong "rank-preservation"/cross-world assumption**)

Traditionally, people made no distinction between the NDE and the CDE, because *under the (highly restrictive) underlying model*, they didn't need too — the *NDE and CDE are the same*!

Problems with the traditional approach

- No interaction term!
- It relies on *strong rank-preservation/cross-world assumption*.
- It assumes *no (uncontrolled) mediator-outcome confounding* (formally, $Y^{a,m} \perp\!\!\!\perp M^a$ given $A = a$) — absence not guaranteed by design in typical trial!

As with traditional methods for time-varying confounding control, *neither conditioning on L nor not conditioning on L works*!



The study of causal **mediation** can be seen as a **special case** of causal inference with **time-varying treatments** ... Rather than having a single treatment that takes different values over time, in mediation analysis we have two different variables — the treatment of interest and the mediator — at different times.

Hernán and Robins, 2020, *Causal Inference: What If*

IPW for inference about controlled directed effects

If, in addition to consistency and positivity, there is *conditional exchangeability* of the form

$$Y^{a,m} \perp\!\!\!\perp A \mid L_0 \text{ and } Y^{a,m} \perp\!\!\!\perp M \mid L_0, L_1, A{=}a$$

(i.e., L_0 blocks all backdoor paths from A to Y, and L_0, L_1, A together block all backdoor paths from from M to Y), then

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where the pseudopopulation is obtained by weighting everyone by $W_0 W_1$ with

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- and $W_1 = M \,/\, \Pr(M = 1 \mid L_0, L_1, A) + (1 - M) \,/\, \Pr(M = 0 \mid L_0, L_1, A)$

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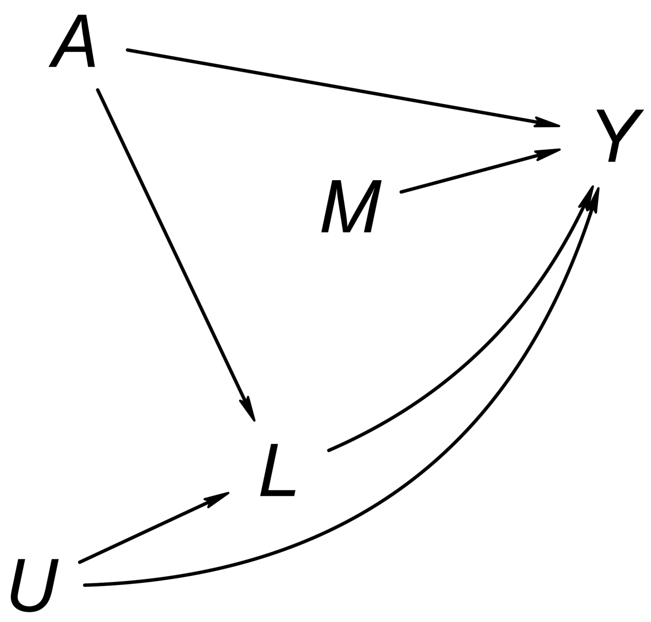

Summary

- We distinguished between *single- and multiple-point interventions* and *static versus dynamic treatment rules*.
- *Mediation* analysis and inference about *time-varying treatments* have *parallels*:
  - Traditional methods fail
  - G-methods overcome issues (but aren't assumption-free!)